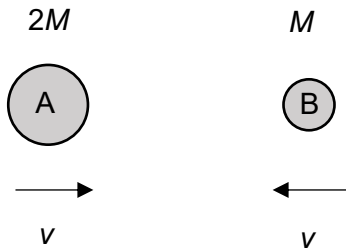


- 8 Two steel balls A and B of masses $2M$ and $1M$ respectively move towards each other with the same speed v and collide elastically.



What are the final velocities of the two balls in terms of v ? Take the rightward direction as positive.

	final velocity of ball A	final velocity of ball B
--	--------------------------	--------------------------

A	$\frac{4}{3}v$	$\frac{7}{3}v$
B	$-\frac{1}{3}v$	$\frac{5}{3}v$
C	$\frac{1}{3}v$	$\frac{2}{3}v$
D	$-v$	$2v$

Ans: B

Let v_1 = final velocity of the ball A

Let v_2 = final velocity of the ball B

By Law of Conservation of Momentum,

$$2Mv + M(-v) = 2Mv_1 + Mv_2$$

$$2v_1 + v_2 = v \text{ ---(1)}$$

Since the collision is elastic,

initial velocity of B relative to A = -(final velocity of B relative to A)

$$(-v) - v = -(v_2 - v_1)$$

$$v_2 = v_1 + 2v \text{ ---(2)}$$

Sub. into (1):

$$2v_1 + (v_1 + 2v) = v$$

$$v_1 = -\frac{1}{3}v$$

Sub. into (2):

$$v_2 = -\frac{1}{3}v + 2v = \frac{5}{3}v$$